

EDUCATION

2010 – 2016	PhD in Experimental Particle Physics, University of Victoria, Canada Supervisors: Rob McPherson & Michel Lefebvre GPA: 8.63/9 Based at CERN (2012) for authorship qualification, ATLAS Experiment
2005 – 2010	Master of Engineering in Applied Physics, INSA Toulouse, France 4 th year: exchange semester on Fundamental Physics (in German), Humboldt-Universität Berlin 5 th year: specialization in Nanophysics & Instrumentation Rank: top 5%

APPOINTMENTS

2023 – 2025	Academic Director, <i>AI for Science</i>*, AIMS South Africa Resident Researcher in Machine Learning, AIMS South Africa * MSc program funded by Google DeepMind: ai.aims.ac.za
2022 – 2023	Visiting Scholar, AIMS South Africa McDonald Institute Ambassador Canada's astroparticle physics network; sponsored fellow
2019 – 2023	Assistant Professor, York University, Canada , jointly appointed as Research Scientist, Fermilab, USA ATLAS Experiment, CERN & Deep Underground Neutrino Experiment, Fermilab
2016 – 2019	Postdoctoral Researcher, DESY, Hamburg, Germany ATLAS Experiment, CERN
2010 – 2016	Graduate Research Associate, University of Victoria, Canada ATLAS Experiment, CERN; based at TRIUMF

PROFESSIONAL DEVELOPMENT

2026	Technical AI Safety, Blue Dot Impact 6-week course with weekly facilitated discussions; certificate of completion
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SKILLS

Research expertise: collider physics, machine learning, advanced data analysis, statistical inference

Programming: Python (NumPy, PyTorch, scikit-learn), C, C++, ROOT, LabVIEW, Unix shell

Communication: university-level instruction, public speaking, science popularization

Mentoring: academic coaching, student placement across international institutions

Languages: French (native), English (fluent), German (fluent), Spanish (highly proficient)

RESEARCH

Highlights

2020 – 2022	Co-convenor of Training & Documentation, DUNE Computing Consortium Deep Underground Neutrino Experiment, Fermilab; arXiv:2310.07342
2019	Co-founded DUNE-Canada group Deep Underground Neutrino Experiment, Fermilab
2018	Observation of the Higgs boson production mode with a top-antitop quark pair ATLAS Experiment, CERN LHC; Phys.Lett. B 784 (2018) 173
2016 – 2019	Led thermal validation and radiation length measurements of the ATLAS ITk silicon strip sensor support structure ATLAS Experiment, CERN LHC; ATLAS-TDR-025
2015 – 2016	pMSSM reinterpretation of LHC Run 1 SUSY searches; featured in CERN Courier ATLAS Experiment, CERN LHC; JHEP 10 (2015) 134

Current Projects

2025 – 2026	Automatic mapping of confidence sets to confidence intervals for intractable statistical models With Prof. Harrison B. Prosper (Florida State U.) & Prof. Olivia Prosper-Feldman (U. of Tennessee) Expected completion: December 2026
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As Resident Researcher in Machine Learning at AIMS South Africa

2025 – 2026	Physics-Informed Neural Network as solver of the photon orbit equation around a black hole With Tlotlo Oepeng (formerly AIMS South Africa) & Prof. H. B. Prosper (FSU) Paper submitted to the Machine Learning for Physical Science (ML4PS) 2026 workshop
2025	Graph autoencoders for anomaly detection in particle physics With Raktim Mukherjee (Universität Heidelberg), Prof. H. B. Prosper (FSU)

As Research Scientist at Fermilab (York University, Toronto, Canada)

2020 – 2023	DUNE Computing Consortium Led first R&D on an interactive analysis facility for DUNE using columnar computing Re-designed and co-organized three DUNE Computing tutorials → Front. Big Data 8 (2025) 1497622 Modernized DUNE Computing wiki; designed & deployed DUNE's automated glossary
2019 – 2023	DUNE-Canada group Secured 3-year NSERC funding to launch first Canadian effort (3 PIs) Secured 5-year CFI infrastructure funding for a cryogenic facility for ArgonCube R&D

As Postdoctoral Researcher at DESY, Hamburg, Germany

2017 – 2019	Observation of $t\bar{t}h$ ($\rightarrow b\bar{b}$) process, ATLAS Experiment, CERN LHC Developed interface for Matrix Element Method in official analysis code Validated Boosted Decision Trees' input distributions across Monte Carlo generators Led fake lepton background estimate, one-lepton channel → Phys.Lett. B 784 (2018) 173
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2016–2019	ATLAS Inner Tracker upgrade for High-Luminosity LHC Led a team (3 engineers, 3 technicians, 4 students) to deliver first measurements of the prototype's thermal performance; developed infrared camera data acquisition interface Led first measurements of radiation length (X/X_0) using EUDET telescopes Created custom entries of prototype's components in the production database
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As Graduate Student Research Associate, University of Victoria, Canada

2013–2015	Search for electroweak Supersymmetry, ATLAS Experiment, CERN LHC Main analyzer and editor of internal note of the search in the $h \rightarrow b\bar{b}$ channel Reinterpreted the search as part of Run 1 summary paper in the context of the 19-parameter pMSSM JHEP 10 (2015) 134
2012–2013	ATLAS Liquid Argon Calorimeter Operation Software on-call expert during 2012 data taking (leading to the Higgs boson discovery) Co-organized software expert training; created web-portal for shifters in control room Developed ATLAS Online Software plugin revision to filter error messages

ENGINEERING EXPERIENCE

2010 5 months	ATLAS Forward Calorimeter upgrade, TRIUMF, Vancouver, Canada Instrumented a radiation hardness test for diamond sensors using TRIUMF's cyclotron proton beam; created acquisition interface with TestPoint Developed and tested the data analysis framework with ROOT → JINST 6 P05011 (2011)
2009 3 months	Pulse-shape modelling for IceCube, DESY, Berlin, Germany Developed data analysis framework to measure the transfer function of IceCube Neutrino Observatory's photomultiplier modules, improving detector simulations
2008 2 months	Afterglow tests monitoring, Hilger Crystals Company*, UK Monitored scintillation crystals afterglow tests under X-Rays; created a PhP/MySQL database for managing the test results * developed crystals for the BABAR experiment
2007 1 month	French National High Magnetic Field Laboratory, Toulouse, France Experimentations on the scotch-tape method for graphene exfoliation

RESEARCH SUPERVISION

At AIMS South Africa

2024–2026	Postdoc Ryan Atkin , AIMS South Africa Main supervisor; anomaly detection for ATLAS High-Level Triggers
2025 2 months	MSc Raktim Mukherjee , visitor from Universität Heidelberg, Germany Supervisor; investigation of Graph Auto-Encoder for anomaly detection for ATLAS triggers

At York University

2021–2023	Postdoc Tejin Cai , York University Main supervisor; interactive analysis facility for DUNE on Compute Canada cloud Now: software engineer at Black Duck
2020	MSc Alfred Ferwerda , York University Thesis committee member; CERN ALPHA Experiment

2019 – 2020	PhD Darij Starko , York University Local supervisor; ATLAS qualification task, Flavour tagging group, CERN Now: data science consultant at Wyatt AI Inc.
2021	BSc Amin Jafarov , York University, NSERC USRA recipient Main supervisor; searches for long-lived particles using ML Now: Medical School, University of Central Lancashire, UK
2021	BSc Anita Nasiri , York University Co-op research project supervisor; long-lived particle event 3D visualisation in ATLAS
2020	BSc Majd Salam , York University Co-op research project supervisor; first ML classifier for ATLAS long-lived particle search Now: PhD candidate, Theoretical Astrophysics, Western University, Canada
2020	BSc Damith Deshan , York University Co-op research project supervisor; automatic scans of chip pack labels for ATLAS ITk production database

As postdoctoral researcher at DESY

2016 – 2019	PhD Jan-Hendrik Arling , DESY Local technical supervisor; ATLAS ITk upgrade Now: junior staff at DESY
2018	BSc Anna Beck , DESY summer student Supervisor; ATLAS ITk thermal characterization PhD at University of Warwick; now postdoctoral researcher at MIT
2017	BSc Maximilian Caspar , DESY summer student Supervisor; ATLAS ITk upgrade, instrumentation PhD at Universität Wuppertal/DESY; now consultant at Basycon
2016 – 2017	PhD Yasiel Delabat Diaz , DESY Local supervisor; ATLAS qualification task, ITk upgrade PhD at Humboldt-Universität/DESY; now software developer at Safetec

As research assistant at University of Victoria

2011	BSc Philipp Koenig , University of Victoria Co-op project co-supervisor; ATLAS Higgs group PhD at Universität Bonn; now senior data analyst at Syna
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TEACHING EXPERIENCE

Invited Lectures

Main course website: clairedavid.github.io/intro_to_ml

Oct. 2025	Introduction to Artificial Intelligence & Data Science, AIMS Cameroon Graduate course; 15 lectures (30 hours) + tutorials (10 hours); 108 students Skill course, required for all MSc streams
Sep. 2025	Introduction to Machine Learning (modernized version), AIMS South Africa MSc course <i>AI for Science</i> ; 10 lectures (20 hours) + tutorials (5 hours); 33 students Co-teaching with Prof. Harrison Prosper (FSU)

Jul. 2025	Basic Concepts in Machine Learning, EASY-ML* Summer School, Nairobi, Kenya 10 lectures (15 hours) + tutorials (5 hours); 35 students * East-African School for Young Researchers on Advanced Machine Learning; CNRS, France
Sep. 2024	Introduction to Machine Learning, AIMS South Africa MSc course <i>AI for Science</i> ; 15 lectures (30 hours) + tutorials (10 hours); 21 students Delivered again in December–January for 10 students who joined late due to visa delays
Apr. 2024	Machine Learning by Hand, AIMS South Africa MSc course <i>AI for Science</i> ; 15 lectures (30 hours) + tutorials (10 hours); 14 students Adapted content for students joining in January due to visa delays
Feb. 2024	Introduction to Artificial Intelligence & Data Science, AIMS Cameroon Graduate course; 15 lectures (30 hours) + tutorials (10 hours); 35 students
Jan. 2024	Introduction to Machine Learning, AIMS Senegal Graduate course; 15 lectures (30 hours) + tutorials (10 hours); 20 students
Sep. 2023	Introduction to Machine Learning, AIMS South Africa MSc course <i>AI for Science</i> ; 15 lectures (30 hours) + tutorials (10 hours); 16 students + 24 auditing students from the Mathematical Science stream
May 2023	Introduction to Machine Learning & Data Science, AIMS Ghana Graduate course; 15 lectures (30 hours) + tutorials (10 hours); 19 students
Mar. 2023	Learn the Math behind Neural Networks, University of Johannesburg, South Africa One-day workshop: theory (3 hours) and practice in Python (3 hours); 22 participants Open to all academic levels from 4 th year undergraduate
Feb. 2023	Machine Learning in Particle Physics, AIMS South Africa Graduate course; 15 lectures (30 hours) + tutorials (10 hours); 33 students
Jan. 2023	Machine Learning in Particle Physics, AIMS Senegal Graduate course; 15 lectures (30 hours) + tutorials (10 hours); 19 students
Dec. 2022	Introduction to Machine Learning, ASP2022*, Gqeberha, South Africa Two morning sessions (6 hours); 40 participants * African School of Physics: africanschoolofphysics.org
Jan. 2019	Life of a Particle: Stochastic Modelling in Particle Physics, AIMS Ghana Graduate course; 15 lectures (30 hours) + tutorials (10 hours); 38 students Co-taught with Dr. Samuel Meehan

Teaching at York University

Fall 2021	Special Relativity and Introduction to Modern Physics (PHYS2040) Second year undergraduates in Physics & Engineering Majors (online); 57 students
Fall 2020	Special Relativity and Introduction to Modern Physics (PHYS2040) Second year undergraduates in Physics & Engineering Majors (online); 64 students
Fall 2019	Special Relativity and Introduction to Modern Physics (PHYS2040) Second year undergraduates in Physics & Engineering Majors; 58 students

Previous Teaching Experience

Sep. 2018	Autumn School for High Energy Physics, Maria Laach, Germany Invited group coordinator; tutoring graduate students projects in experimental particle physics; 15 participants
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Jan. 2016	TRIUMF Junior Research School, Vancouver, Canada Invited lecturer: “The Higgs boson mechanism”; 10 Singaporean college students
Jan. 2015	TRIUMF Junior Research School, Vancouver, Canada Invited lecturer: “The Higgs boson mechanism”; 10 Singaporean college students

STUDENT PLACEMENT

As Academic Director at AIMS South Africa

Matched students with capstone supervisors; one-on-one mentoring & tailored career guidance; connected and referred students for post-AIMS positions.

	33 students guided to MSc graduation, <i>AI for Science</i> Selected placements: Ahmed Nadir , Research Intern, ML & Global Health Network, University of Oxford; active learning for survey design Antony Gitau , PhD student, University of South-Eastern Norway; AI for medical imaging Benedict Ejelonu , G-RIPS Fellow, Tohoku University, Japan; AIMS–Tübingen Data Science & AI Fellow Harlette Denebeye , Marie Skłodowska-Curie Doctoral Researcher, Université Paris-Saclay/CEA-CNRGH; trustworthy AI Josiah Kimani , AI for Weather Prediction researcher, AfriClimate AI; 2 nd prize poster, Deep Learning Indaba 2025 Koomi Toussaint Amoussouvi , AI for Weather Prediction researcher, AfriClimate AI; 1 st prize poster, Deep Learning Indaba 2025 Mushtaha El-Amin , Visiting Researcher, LiGHT Laboratory, EPFL; global health & humanitarian response technologies Omer Ebead , Research Fellow, Cooperative AI Foundation; multi-agent privacy; intern at InstaDeep on MARL Onintsoa Rakotondranisa , PhD student, University of Cape Town; astrophysics Rodeo Toha , Head of data and AI at Grow-UP AI, Abomey-Calavi, Benin Tim Hitge , Research Assistant, Imperial College London; surrogate models & simulation-based inference; paper + poster at NeurIPS 2025
2024 – 2025	
	30 students guided to MSc graduation, <i>AI for Science</i> inaugural cohort Selected placements: Abel Legese , Senior AI Research Engineer, Ersilia; open-source drug discovery Carlin Foka , PhD student, Centre for Epidemic Response & Innovation (CERI) Catherine Essuman , PhD candidate (from Sept. 2026), Ontario Tech University, Canada; Canada Impact+ Research Award recipient Daniel Rajaonarivelomanantsoa , Research Fellow, InstaDeep; first Malagasy oral presenter at NeurIPS 2025 ; PhD, Stellenbosch University Dariose Diffo , PhD student, Università di Bologna; AI & computational methods for astronomy and physics Haingo Andry , Recipient of the 2025 Google PhD Fellowship, Stellenbosch University/CERI Herman Tesso , Doctoral Researcher, Aalto University, Finland; first-author paper at AABI workshop, ICLR 2025; simulation-based inference & spatial statistics Isaac Omolayo , PhD student, Universität Tübingen; contrastive learning for connectome-constrained neural networks, fruit fly visual system Shâma Wache , PhD student, TU Chemnitz, Germany; inverse problems in medical imaging Tiana Ramonjison , MSc student, Simon Fraser University/TRIUMF; dark matter detector design
2023 – 2024	

Informal Mentoring & Connections

2025	Catherine Essuman & Pelumi Aderinto , <i>AI for Science</i> alumni Led on my behalf the tutorial “The Math Behind Neural Networks” at Deep Learning Indaba, Lagos, Nigeria
2025	Angel Bashyal , MSc in Astrophysics, Tribhuvan University, Nepal Met at BCVSPIN 2024; connected to potential supervisors via McDonald Institute network Recipient of Swiss Government Excellence Scholarship; PhD in Astrophysics, start Sept. 2026
2025	Nishan Lamichhane , MSc student, Tribhuvan Vishwavidalaya, Nepal Connected with Prof. Aaron Vincent, Queen’s University, for honour’s research project Now accepted for a PhD in Physics & Astronomy at Queen’s University; start Sept. 2026
2024	Jeffrey Katabo , MSc student, AIMS Cameroon Connected with Prof. Joel Zylberberg (CIFAR Fellow, York University); meta-learning for models of primate visual cortex (AIMS capstone project) Now pre-doctoral research fellow, Freie Universität Berlin / Bernstein Center for Computational Neuroscience
2023	Ufuk Çakır , AIMS–Universität Heidelberg research intern DPhil student, Intelligent Earth CDT, University of Oxford; RL for wildfire response

ACADEMIC SERVICE

Service to AIMS

2024–2026	Designed and led student onboarding programme, AIMS South Africa Week-long induction: campus life, values, systems, cultural exchange, wellness walks 4 editions: Sept. 2024, Jan. 2025, Sept. 2025, Jan. 2026
2023–2025	Founded student clubs to foster initiative and campus life, AIMS South Africa Student-run, with dedicated budget; adapted each year to cohort interests 2023–24: Arts, Fitness, Hike & Walk, Paper Reading, Running 2024–25: Board Game, Cultural, Dance, Media, Running, Soccer
2023–2025	Co-created formal lecturer contracts and tutor agreements, AIMS South Africa

Service to Fermilab/York University

2020–2021	Member, Fermilab Neutrino Division postdoc hiring committee
2020–2021	Co-chair, Physics & Astronomy Department Weekly Colloquium, York University
2019–2020	Member, Faculty of Science Award Committee, York University

Service to the Field

2024	Session co-chair, Sustainability Track, 42nd International Conference on High Energy Physics (ICHEP), Prague First dedicated sustainability track at ICHEP; environmental impact of HEP accelerators, detectors and computing
2022–2023	Sustainable HECAP+ Initiative Creator of sustainable-hecap-plus.github.io ; advocate for sustainable practices in high-energy physics
2022	Judge, Best Student Competition, Particle Physics division Canadian Association of Physicists’ Congress, Hamilton, Canada

2021 – 2022	Elected Vice-Chair, Particle Physics Division, Canadian Association of Physicists (CAP)
2021	Session convener, Neutrino session, CAP Congress 2021
2020 – 2022	Co-convener, DUNE Training & Documentation working group, Fermilab
2019 – 2020	Institute contact (York University), ATLAS Inner Tracker production database

Memberships

2024 – 2025	Hydrogen Intensity and Real-time Analysis eXperiment (HIRAX) Member of Management Team
2023 – 2026	South African Institute of Physics (SAIP)
2020 – 2022	Canadian Association of Physicists (CAP)
2019 – 2022	Institute of Particle Physics (IPP), Canada

Peer Review

2026	Reviewer, Nederlandse Organisatie voor Wetenschappelijk Onderzoek (NOW) Dutch Research Council
2025	Nominated reviewer, Machine Learning for Physical Sciences Workshop (ML4PS), NeurIPS 2025 Reviewed 2 papers
2024	Nominated reviewer, Machine Learning for Physical Sciences Workshop (ML4PS), NeurIPS 2024 Reviewed 2 papers
2019	Referee, Journal of Instrumentation Nominated by peers
2017	Referee, Journal of Instrumentation Nominated by peers

Editorial Work

2021	Selected reviewer, DUNE Vertical Drift Conceptual Design Report Chapter: Trigger & Data Acquisition
2019 – 2021	Editorial Board Member, ATLAS Collaboration “Configuration and performance of the ATLAS b-jet triggers in Run 2”; Eur. Phys. J. C 81 (2021) 1087

Event Organization

2025	International Masterclass “Hands-on Particle Physics”, AIMS South Africa First edition hosted by AIMS; 25 high school students from 8 Cape Town schools Organized in partnership with IPPOG; live conference with CERN
2023 – 2024	Siyakhula Festival, AIMS 20th Birthday Celebration, Muizenberg, South Africa Organizing committee member; 6-day festival, March 17–22, 2024
2021	DUNE Computing Tutorial, extended version (online) Co-organizer; 3-day format with interactive platform, quizzes and expert Q&A

2021	Poster session organizer, TIPP 2021 (online) International Conference on Technology and Instrumentation in Particle Physics; GatherTown platform
2021	DUNE Computing Tutorial, one-day workshop (online) Main organizer; teaching basics of DUNE computing to new members
2019	ATLAS Higgs & Top Quark Workshop (HTop), DESY, Hamburg, Germany Member of the Local Organizing Committee

Other Service

2007 – 2008	Founder & spokesperson, Commission Réussite, INSA Toulouse Designed & implemented programme where students in difficulty choose a mentor among professors Approved by administration as a dedicated commission; system still active in 2026
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GRANTS & AWARDS

Research Grants

2022 – 2027	Canadian Foundation for Innovation (CFI) John R. Evans Leaders Fund (JELF) (PI, CAD \$125,000) Next-generation neutrino detector for DUNE
2022 – 2023	Compute Canada Fast Track Competition, one-year extension (PI, CAD \$23,339)
2021 – 2022	Compute Canada Resources Competition (PI, CAD \$32,225) Interactive Analysis Facility for DUNE
2021 – 2022	NSERC Discovery COVID-19 Extension (non-competitive, CAD \$50,000)
2020 – 2023	NSERC Subatomic Physics Discovery Grant (Co-PI, CAD \$465,000) Neutrinos exposed: from the cosmos to deep underground
2020 – 2021	NSERC Subatomic Physics Discovery Grant (PI, CAD \$50,000) Higgs boson and top quark in ATLAS: gateway to new physics
2019 – 2023	York University Internal Research Fund, Startup (CAD \$110,000)

Awards & Recognitions

2025	Breakthrough Prize in Fundamental Physics Shared with ATLAS, CMS, ALICE and LHCb collaborations (13,508 researchers) breakthroughprize.org
2020	York University Minor Research Grant & Junior Faculty Fund (CAD \$3,100)
2017 – 2018	DESY Bonus Merit (€3,200) Thermal validation of ATLAS ITk prototype
2016 – 2017	DESY Bonus Merit (€3,200) First infrared measurements of ATLAS ITk sensor structures
2016	DESY Postdoctoral Fellowship , High Energy Physics division
2011 – 2012	Nora and Mark de Goutiere Memorial Fellowship (CAD \$10,000)
2010	University of Victoria Entrance Award (CAD \$10,000) Nominated for Vanier Scholarship, Dept. of Physics & Astronomy, UVic

PRESENTATIONS

International Conferences

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| 2026 | <p>High Dimensional Problems for Statistical Methods in Fundamental Physics, BIRS, Banff, Canada</p> <p>Invited talk: “Leveraging Machine Learning, and its limits” (co-presenter: Anthony Ozerov)</p> |
| 2025 | <p>32nd International Workshop on Deep Inelastic Scattering (DIS2025), Cape Town, South Africa</p> <p>Invited plenary talk: “Diversity, Equity and Inclusion in Physics”</p> |
| 2023 | <p>11th Large Hadron Collider Physics Conference (LHCP), Belgrade, Serbia</p> <p>Invited talk: “Statistics on participation of emerging countries in LHC and other experiments”</p> |
| 2020 | <p>30th International Linear Accelerator Conference (LINAC), Liverpool, UK (online)</p> <p>Invited panelist: “Women in Science and Engineering: Should I stay or should I go now?”</p> |
| 2019 | <p>Conference on Flavor Physics and CP Violation (FPCP), Victoria, Canada</p> <p>Invited talk: “Higgs boson couplings to quarks at the ATLAS experiment”</p> |
| 2018 | <p>Conference on Standard Model at LHC, Berlin, Germany</p> <p>Young Scientist talk: “Search for $t\bar{t}H$ production in the $H \rightarrow b\bar{b}$ channel”</p> |
| 2017 | <p>11th International Conference on Position Sensitive Detectors (PSD11), Milton Keynes, UK</p> <p>Invited plenary talk: “A new strips tracker for the upgraded ATLAS ITk detector”</p> |
| 2016 | <p>Lake Louise Winter Institute, Alberta, Canada</p> <p>Invited plenary talk: “Search for supersymmetry with the ATLAS detector in final states with leptons, jets and missing transverse momentum”</p> |
| 2014 | <p>International Conference on High-Energy Physics (ICHEP), Valencia, Spain</p> <p>Poster: “Search for chargino and neutralino production with a Higgs boson in the decay chain with one or three leptons final state events”</p> |
| 2013 | <p>International Conference on New Frontiers in Physics (ICNFP), Kolymbari, Greece</p> <p>Invited talk: “ATLAS results in SUSY searches for 3rd generation squarks and electroweak production”</p> |

Funding Agencies & Institutional Reviews

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| 2021 | <p>Institute of Particle Physics (IPP) Annual General Meeting (online)</p> <p>Invited talk: “Deep Underground Neutrino Experiment: current status, Canadian effort & funding”</p> |
| 2018 | <p>DESY Program-Oriented Funding (POF) Evaluation, Hamburg, Germany</p> <p>Invited poster: “ATLAS and CMS tracker end-caps for the High-Luminosity LHC” (co-author Dr. J. Keaveney); official tour guide for Helmholtz Association reviewers</p> |
| 2016 | <p>Helmholtz Programme Matter and the Universe, Mainz, Germany</p> <p>Invited poster: “Anatomy of ATLAS and CMS strip tracker end-caps for High-Luminosity LHC”</p> |
| 2013 | <p>TRIUMF International Review Committee, Canada</p> <p>Invited poster: “Search for Higgs production in supersymmetric decay cascades with ATLAS”</p> |
| 2013 | <p>VIP Visit, ATLAS Control Room, CERN, Geneva, Switzerland</p> <p>Invited speaker for P. Young (Chair, TRIUMF Board of Management) & G. Patry (CEO, Canada Foundation for Innovation)</p> |

Local & Regional Conferences

2021	Canadian Association of Physicists Congress (CAP), online Invited talk: “Deep Underground Neutrino Experiment: Review and Recent Progress”
2021	Third Summer Particle Astrophysics Workshop, Queen’s University, Canada (online) Plenary talk: “The Deep Underground Neutrino Experiment: DUNE”
2020	Winter Nuclear & Particle Physics Conference (WNPPC), Banff, Canada Plenary talk: “Higgs boson physics and upgrade work with ATLAS at the Large Hadron Collider”
2019	HTop Workshop, DESY, Hamburg, Germany Plenary talk: “Small backgrounds in the leptonic channels”
2018	Preliminary Design Review ITk Strip Local Support Structures, CERN Final presentation of results on thermo-mechanical petal prototypes; passing review
2018	Forum on Tracking Detector Mechanics, Valencia, Spain Plenary talk: “Thermo-mechanical characterization of Petals as Local Support Structures for the ATLAS ITk Strip Detector”
2018	HTop Workshop, CERN, Switzerland Invited talk: “Fakes and other backgrounds in $t\bar{t}H(b\bar{b})$ leptonic channels” (co-author J. Mellenthin)
2014	LHCC, CERN, Switzerland Poster: “Search for charginos, neutralinos and sleptons in one or two leptons final state events with ATLAS” (co-author R. Ueno)
2011	ATLAS Canada Workshop, Vancouver, Canada Talk: “Search for SUSY Higgs in the $b\bar{b}$ final state”

Public Talks

2026	Career Thursday, Qhubeka Youth Development Programme, Langa, South Africa Featured guest; Q&A on experimental physics & AI in education; organized by Just Grace
2023	Université Kongo, Mbanza-Ngungu, Congo DRC Co-talk with Dr. Nour Skaf: “From the smallest to the largest: a journey from particle physics to astrophysics”
2023	Dakar Academy, Senegal Invited speaker: experimental particle physics to 200+ college-preparatory students
2022	Neutrino Day, Sanford Underground Research Facility, Lead, South Dakota, USA Keynote speaker at Lead’s Opera House; VIP underground visit with YouTuber Physics Girl (Dianna Covern) Live radio broadcast for South Dakota radio about DUNE and Neutrino Day
2021	Royal Astronomical Society of Canada, Toronto, Canada Invited speaker, Speaker’s Night livestream: “What is DUNE? Embark on the underground telescope to pierce the secrets of matter!” [YouTube]
2015	Science Innovation & Technology Speaker Series, Richmond, BC, Canada “The Triumphant Discovery of the Higgs Boson”
2014	Cafés Scientifiques, Alliance Française, Vancouver, Canada “La triomphante découverte du boson de Higgs”; organized by the French Consulate

Invited Colloquia

2026	Integrative Think Tank (ITT), AIMS South Africa Five-day industry-academia workshop; invited talk: “How I became a scientist: the backstory not read on my CV”
2025	AIMS Cameroon, Limbé, Cameroon “A journey inside matter: experimental particle physics with ATLAS and DUNE”
2025	AIMS Cameroon, Limbé, Cameroon “How I became a scientist: the backstory not read on my CV”
2024	AIMS Cameroon, Limbé, Cameroon “A journey inside matter: experimental particle physics with ATLAS and DUNE”
2024	AIMS Cameroon, Limbé, Cameroon “How I became a scientist: the backstory not read on my CV”
2023	Université de Montréal, Département de Physique, Canada “Half a year in Africa: AI, AIMS & opportunities”
2023	East African Institute for Fundamental Research (EAIFR), Kigali, Rwanda Co-talk with Dr. Nour Skaf: “From the smallest to the largest: a journey from particle physics to astro-physics”
2023	University of Ghana, Department of Physics, Accra, Ghana “A journey inside matter: experimental particle physics with ATLAS and DUNE”
2023	AIMS Rwanda, Kigali, Rwanda “A journey inside matter: experimental particle physics with ATLAS and DUNE”
2023	AIMS Cameroon, Limbé, Cameroon “A journey inside matter: experimental particle physics with ATLAS and DUNE”
2023	Université de Lomé & CERME, Lomé, Togo “The ATLAS Experiment at CERN: a detector, an analysis and an upgrade program”
2023	International Chair in Mathematical Physics & Applications, Cotonou, Benin “A journey inside matter: experimental particle physics with ATLAS and DUNE”
2023	AIMS South Africa, Muizenberg, South Africa “How I became a scientist: the backstory not read on my CV”
2020	McGill Physical Society Colloquium, Montréal, Canada (online) “Data analysis and detector development with ATLAS at the Large Hadron Collider” [YouTube]
2019	York University, Dept. of Physics & Astronomy Colloquium, Toronto, Canada “Sorting needles in a haystack and cooling a sandwich: data analysis and detector development with ATLAS”

PUBLICATIONS

2026	David C., Oepeng T., Prosper H., “PINN-Slicer: Exploiting Rotational Symmetry to Solve the Photon Orbit Equation Around a Black Hole” Submitted to the ML4PS 2026 workshop
2025	Reinsvold Hall A., Skidmore N., Benelli G., Carlson B., David C. et al., “Training and onboarding initiatives in high energy physics experiments” Front. Big Data 8 (2025) 1497622

2025	Sustainable HECAP+ Initiative, “Environmental sustainability in basic research: a perspective from HECAP+” JINST 20 (2025) P03012 ; www.hecap.eco
2022	DUNE Collaboration, “Design, construction and operation of the ProtoDUNE-SP Liquid Argon TPC” JINST 17 (2022) P01005
2021	ATLAS Collaboration, “Search for exotic decays of the Higgs boson into long-lived particles in pp collisions at $\sqrt{s} = 13$ TeV using displaced vertices in the ATLAS inner detector” JHEP 11 (2021) 229
2021	DUNE Collaboration, “Deep Underground Neutrino Experiment (DUNE) Near Detector Conceptual Design Report” Instruments 5 (2021) 31
2020	Poley L., Sawyer C. et al., “The ABC130 barrel module prototyping programme for the ATLAS strip tracker” JINST 15 (2020) P09004
2020	DUNE Collaboration, “Long-baseline neutrino oscillation physics potential of the DUNE experiment” Eur. Phys. J. C 80 (2020) 978
2018	ATLAS Collaboration, “Observation of Higgs boson production in association with a top quark pair at the LHC with the ATLAS detector” Phys. Lett. B 784 (2018) 173–191
2018	ATLAS Collaboration, “Search for the SM Higgs boson produced in association with top quarks and decaying into a $b\bar{b}$ pair in pp collisions at $\sqrt{s} = 13$ TeV with the ATLAS detector” Phys. Rev. D 97 (2018) 072016
2018	Albertsson K. et al., “Machine Learning in High Energy Physics Community White Paper” J. Phys.: Conf. Ser. 1085 (2018) 022008
2017	ATLAS Collaboration, “Technical Design Report for the ATLAS Inner Tracker Strip Detector” CERN-LHCC-2017-005 ; ATLAS-TDR-025
2015	ATLAS Collaboration, “Summary of the ATLAS experiment’s sensitivity to supersymmetry after LHC Run 1 — interpreted in the phenomenological MSSM” JHEP 10 (2015) 134
2015	ATLAS Collaboration, “Search for direct pair production of a chargino and a neutralino decaying to the 125 GeV Higgs boson in $\sqrt{s} = 8$ TeV pp collisions with the ATLAS detector” Eur. Phys. J. C 75 (2015) 208
2013	ATLAS Collaboration, “Search for chargino and neutralino production in final states with one lepton, two b -jets consistent with a Higgs boson, and missing transverse momentum with the ATLAS detector in 20.3 fb^{-1} of $\sqrt{s} = 8$ TeV pp collisions” ATLAS-CONF-2013-093
2011	Axen D. et al., “Diamond detector irradiation tests at TRIUMF” JINST 6 (2011) P05011

PROCEEDINGS

2025	David C. , “What is AI, what is it not, how we use it in physics and how it impacts. . . you” BCVSPIN 2024: Particle Physics and Cosmology in the Himalayas, Kathmandu, Nepal arXiv:2504.01827
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2024	David C. , “Statistics on participation of emerging and developing countries in LHC and other experiments” PoS(LHCP2023) 450 (2024) 062
2018	David C. , “A new strips tracker for the upgraded ATLAS ITk detector” Proceedings of PSD11 JINST 13 (2018) C01003
2016	David C. , “Search for chargino and neutralino production with a Higgs boson in the decay chain in one or three leptons final state events with ATLAS” Nucl. Part. Phys. Proc. 273–275 (2016) 2400–2402
2014	David C. , “New ATLAS results in SUSY searches for 3 rd generation squarks and electroweak production” EPJ Web of Conferences 71 (2014) 00034

OTHER ACTIVITIES

Panels

2025	SA-CERN 15 Year Celebrations, iThemba LABS, South Africa Invited panelist: “Fundamental Science for South African Development”
2023	Science & Technology Week, Kinshasa, Congo DRC Invited panelist: “Science and technology for sustainable development in Africa” (in French)
2023	11th Large Hadron Collider Physics Conference (LHCP), Belgrade, Serbia Invited panelist: “Challenges for emerging countries to join/participate in LHC experiments”
2020	30th International Linear Accelerator Conference (LINAC), Liverpool, UK (online) Invited panelist: “Women in Science and Engineering: Should I stay or should I go now?”
2014	Screening of “Particle Fever”, Vancouver International Film Festival, Canada Invited panelist with movie producer & Prof. David Kaplan
2011	Emily Carr University of Art and Design, Vancouver, Canada Panel speaker: “But $\hbar$? Particle Interactions in Physics and Arts: Artistic Transformation”

Science Festivals & Tours

2023	Village of Sciences, École de la Gombé, Kinshasa, Congo DRC Introduced CERN and ATLAS to 10,000+ high school students; trained selected learners as booth animators
2017	DESY Open Day (DESYDAY), Hamburg, Germany Organizer of ATLAS exhibition & infrared camera booth; 20,000+ visitors
2015– 2016	TRIUMF Tour Guide, Vancouver, Canada Monthly tours for high schools, graduate students and the general public
2014	ATLAS Masterclass, University of Victoria, Canada One-day hands-on for high-school students; invited talk: “Data analysis with ATLAS at the LHC”; organized by VISPA
2013	TRIUMF Open House, Vancouver, Canada Volunteer at the ATLAS booth; 1,200+ visitors

Online & Media

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| 2023 | ATLAS Virtual Visit, ATLAS Control Room, CERN
Lycée Liziba, Kinshasa, Congo DRC; girls' secondary school [atlas.cern] |
| 2023 | ATLAS Virtual Visit, ATLAS Control Room, CERN
Malaika School, Lubumbashi, Congo DRC; free education for 400+ girls [atlas.cern] |
| 2021 | Allan I. Carswell Observatory, York University, Canada
TeleTube series: "The Triumphant Discovery of the Higgs Boson at the Large Hadron Collider" [YouTube] |
| 2020 | "The Drift" series, Sanford Underground Research Facility, USA
"What is so special about neutrinos?" on SURF's YouTube channel [YouTube] |
| 2020 | Neutrino Day, SURF's Free Science Festival (online)
"DUNE scientists answer your questions" [YouTube] |
| 2013 | ATLAS Virtual Visit, ATLAS Control Room, CERN
Europe Day Symposium, Vancouver; TELUS World of Science Celebration, Vancouver [video] |